

STATISTICAL RESEARCH REPORT (STAT452)

Red Wine Quality & Factorial Experimental Design

L1/L2 Regularized Predictive Modeling & Two-Factor ANOVA Analysis

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📅 Academic Year 2025 – 2026

PART1

Red Wine Quality Prediction

Modeling Sensory Scores from 11 Physicochemical Features with
Strict Leakage Control Protocol

EDA & Data Distribution Characteristics

Vinho Verde Dataset

- **Sample Size:** 1,359 unique records (240 exact duplicate rows removed prior to split).
- **Target Variable (Quality):** Median score from expert wine tasters (observed range: 3 - 8).
- **Locked Split:** Stratified 80% Training (N=1,088) & 20% Holdout (N=271).

Skewness & Transformations

- **Response Density:** Ratings cluster heavily at score 5 (N=437) and score 6 (N=428).
- **Severe Right-Skewness:** Residual Sugar (4.45), Chlorides (5.45), Sulphates (2.53).
- **Preprocessing:** Winsorized at 1st-99th percentiles & `log1p` transformed for tail smoothing.

Correlation & Multicollinearity

Primary Signal Drivers: Alcohol exhibits the strongest positive correlation ($r = 0.475$), while Volatile Acidity has the strongest negative correlation ($r = -0.369$).

⚠️ Multicollinearity Diagnostic

VIF flags **Fixed Acidity at 7.90** and **Density at 6.99**, reflecting strong chemical dependencies across acid and density features.

Strategy: Retain all features to prevent arbitrary selection bias; deploy Regularization Shrinkage (L1/L2) to stabilize variance.

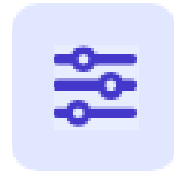


Leakage Control Protocol & Lasso Selection



Leakage Prevention

Holdout set (N=271) locked completely. Winsorization bounds, means, and SDs fit exclusively within Analysis Folds of 5-fold CV.



Baseline Models

Mean Baseline achieved CV RMSE = 0.824. Full 11-feature OLS achieved CV RMSE = 0.672 but exhibited substantial coefficient instability.



Lasso 1-SE Selection

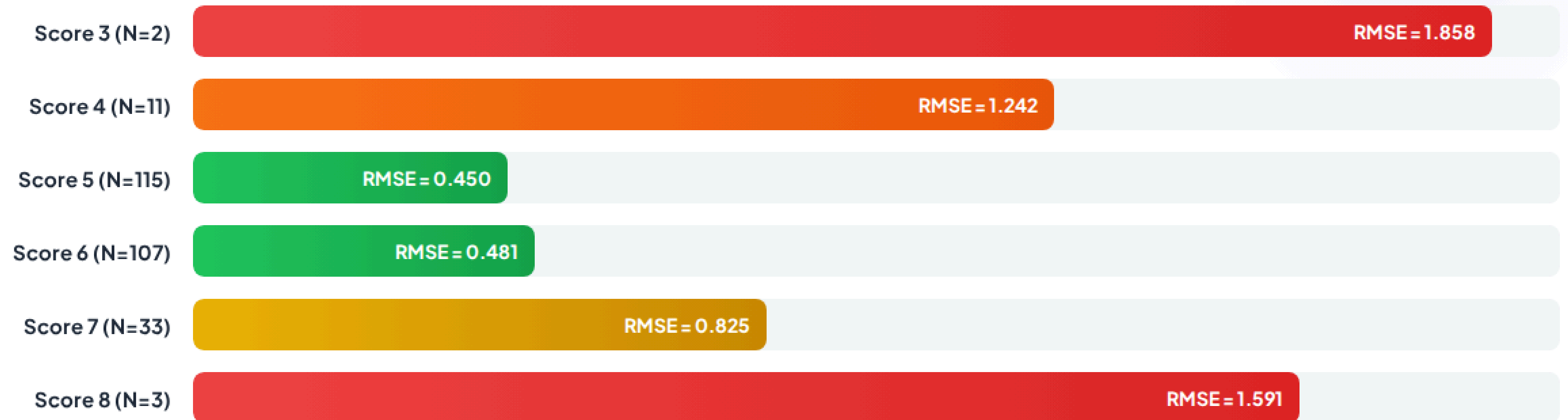
1-SE rule filtered the model down to **7 key predictors**: Alcohol, Volatile Acidity, Sulphates, Chlorides, pH, Total SO₂, Fixed Acidity.

Regularized Regression Model Comparison

Model Architecture	Parameter (α)	CV RMSE	Holdout RMSE	Holdout MAE	Holdout R^2	Lock Status
Mean Benchmark	-	0.8242	0.8198	0.6923	-0.0001	Reference
OLS Baseline	0.00	0.6721	0.6131	0.4803	0.4406	No
Ridge Regression	0.00 ($\lambda=0.047$)	0.6711	0.6134	0.4821	0.4400	CONFIRMED LOCK
Lasso Regression	1.00 ($\lambda=0.005$)	0.6712	0.6139	0.4822	0.4390	No
Elastic Net	0.75 ($\lambda=0.007$)	0.6712	0.6139	0.4822	0.4391	No

Evaluation Summary: Ridge was locked prior to opening holdout based on lowest CV RMSE (0.6711). Generalization test confirmed $R^2 \approx 0.44$, explaining 44% of sensory rating variance

Holdout Error Breakdown by Quality Score



Regression toward the Mean: Predictions are exceptionally accurate for frequent scores (5 and 6), but show systematic error inflation at rare, extreme scores (3, 4, and 8).

PART 2

2x2 Factorial ANOVA Design

Analyzing the Effects of Test Preparation Course & Lunch Category on Student Mathematics Performance

Factorial ANOVA & Effect Sizes

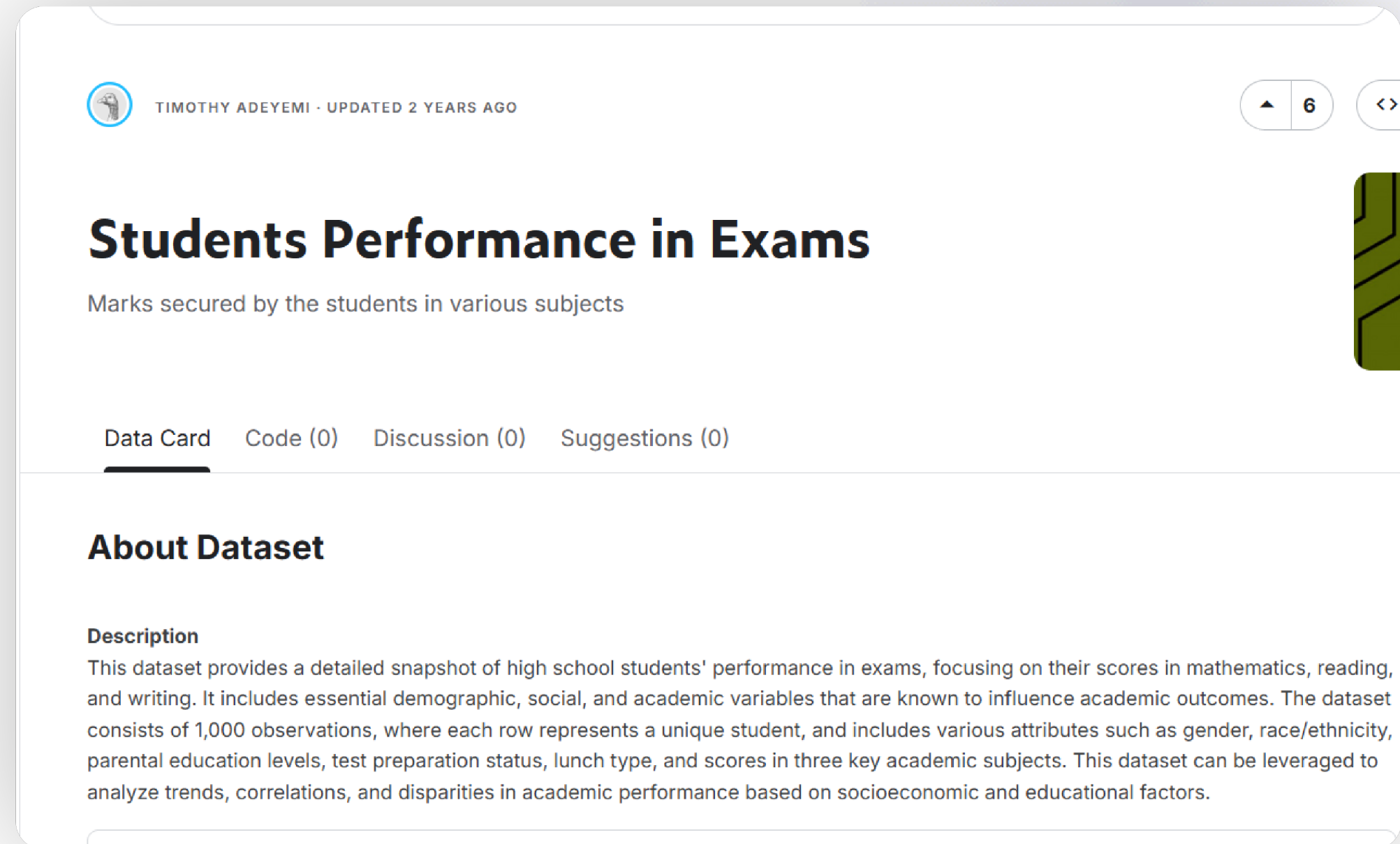
Effect / Source	Point Estimate (95% CI)	F-value	p-value	Partial η^2	Significance Evaluation
Test Preparation Course	+5.97 pts [4.09, 7.85]	38.791	< 0.001	0.037	Significant (Small)
Lunch Category Group	+11.06 pts [9.18, 12.94]	133.132	< 0.001	0.118	Significant (Medium)
Interaction (Prep × Lunch)	-1.14 pts [-4.90, 2.62]	0.352	0.553	0.0004	Not Significant

✔ Dominant Lunch Effect: Socioeconomic proxy impact on Math scores is nearly twice as large as test prep course participation.

ℹ Independent Effects: Test prep benefits are uniform across different lunch socioeconomic categories.

Experimental Design & Student Dataset

- 🎓 **Sample Size:** $N = 1,000$ students (Kaggle Student Performance Dataset).
- 📈 **Dependent Variable (Y):** Math Score (Continuous scale 0 - 100).
- 📚 **Factor A (Preparation):** Test Prep Course (Completed: 358, None: 642).
- 🍴 **Factor B (Lunch):** Lunch Group - Socioeconomic Proxy (Standard: 645, Free/Reduced: 355).



Diagnostics & Follow-up Comparisons

-  **Shapiro-Wilk Test ($W = 0.996$, $p = 0.005$):** Detects mild non-normality due to score boundaries [0, 100]. Large sample size ($N=1,000$) maintains ANOVA robustness.
-  **Brown-Forsythe Test ($F = 1.06$, $p = 0.364$):** Confirms homogeneity of variance across treatment groups.
-  **Holm-adjusted Simple Effects:** Prep effect is +6.54 pts in Free/Reduced group and +5.40 pts in Standard Lunch group (both $p < 0.001$).
-  **Future Experimental Design:** Recommend Randomized Controlled Trials (RCT) controlling for baseline pre-test scores and school/classroom clusters.



Questions & Discussion

Rigorous Statistical Regularization & Experimental Design for
Data-Driven Decision Making.

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